

RENTZOO GO - TRIP COUNTER

Smart Fleet Trip Verification & Fuel Accounting Management System

1. Executive Summary

Rentzoo Go (Trip Counter) is a purpose-built, mobile-first Progressive Web Application (PWA) designed to solve fleet visibility, trip count discrepancy, and fuel leakage challenges in commercial transport and logistics operations. It replaces error-prone paper logbooks with a real-time, tamper-resistant system connecting **Drivers**, **Site Operations Admins**, and **Super Administrators** in a single synchronized platform.

2. Key Challenges & Core Value Proposition

Operational Challenge	Rentzoo Go Solution	Business Impact
Disputed trip counts between drivers and site managers	One-tap verified logging with idempotency keys & dispute audit trails	Eliminates billing disputes & guarantees accurate payroll
Unmonitored diesel theft and unverified fuel refills	Dedicated Diesel Tracking Module linking driver, vehicle, litres, & timestamps	15-25% reduction in unexplained fuel losses
Poor mobile connectivity in transit areas	Offline-capable PWA architecture with client-side synchronization	Zero data loss even in low/no connectivity zones
Lack of centralized management for multi-shift vehicles	Two-slot driver-to-vehicle assignment with shift boundary checks	Optimized fleet utilization and transparent asset accountability

3. Core Application Modules & User Roles

Module / Role	Key Features & Capabilities
Driver Portal (Mobile PWA)	- Minimal, high-contrast UI optimized for one-handed in-cab use - Instant trip logging with duplicate submission protection (idempotency) - Personal fuel logbook with daily, monthly, and aggregate litre metrics - Real-time notification inbox for trip adjustments and fuel record alerts
Admin Operations Panel (Desktop / Tablet)	- Fleet dashboard displaying Reported vs Adjusted vs Verified trips - Single-click verification and reason-coded trip adjustments (+/-) - Scoped Diesel Entry Logger restricted strictly to assigned fleet vehicles - Date-range analytics (Today, Yesterday, Week, Month, Custom)
Super Admin Console (Enterprise Control)	- Complete vehicle lifecycle tracking (Active, Breakdown, Inactive) - Multi-tenant Admin-to-Vehicle and Driver-to-Vehicle shift assignment - Immutable security audit trail logging every login, trip edit, and fuel entry - User provisioning and credential management with RBAC

4. Modern Technology Stack

- **Frontend & App Layer:** Next.js 16 (App Router), React 19, Tailwind CSS v4, Progressive Web App (PWA v2).
- **Backend & Data Layer:** Node.js Route Handlers, Drizzle ORM, PostgreSQL with UUID keys and relational constraints.
- **Security & Integrity:** HTTP-only JWT sessions, BCrypt password hashing, Strict Role-Based Access Control (RBAC), and Timezone-aware date boundaries (Asia/Kolkata default).

5. Implementation Roadmap & Expected ROI

Phase	Deliverables	Timeline
Phase 1: Foundation	Core PWA trip logging, diesel ledger, role-based dashboards, and audit trails	Completed (Active)
Phase 2: Mobile Hardening	Safari/iOS PWA v2 caching fix, SQL-level permission scoping, driver notification alerts	Current Release

Phase 3: Scale & Analytics	Automated mileage/km/litre efficiency analytics, driver dispute flows, and ERP exports	Next Sprint
----------------------------	--	-------------

Conclusion: Rentzoo Go provides a turnkey, high-reliability operational backbone for modern fleet logistics. By bridging mobile driver workflows with administrative oversight and stringent fuel audits, it ensures transparency, accountability, and rapid cost recovery across all vehicle operations.